

Computational Geophysics: What is the best strategy for my problem?

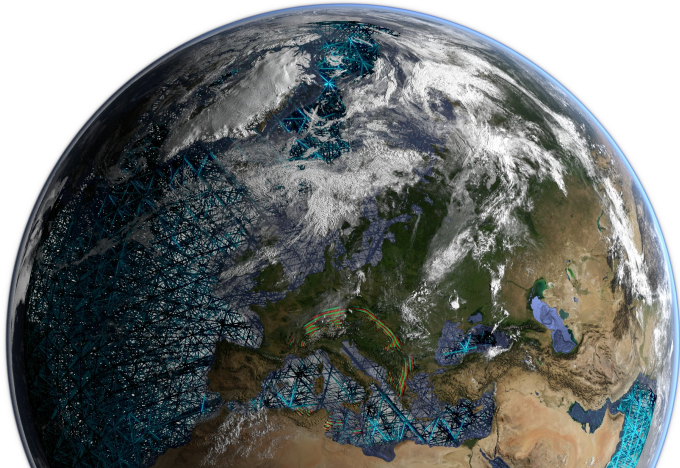
Short Course

Heiner Igel

August 24, 2026

Department of Earth and Environmental Sciences
Ludwig-Maximilians-University Munich

Introduction



Course Content

- Part I
 - Introduction - **Motivation**
 - **Fundamentals** of wave propagation (**wave equations, analytical solutions, reciprocity, superposition principle, dispersion, homogenization**)
- Part II
 - The **finite-difference** method
 - The **pseudospectral** method
 - Linear **finite-element** method
- Part III
 - The **spectral-element** method
 - The **finite-volume** method
 - The **discontinuous Galerkin** method
 - Outlook

What you get ...

- Basic ideas of **discretization**, **size** of a problem, designing a solution
- The fundamentals of **wave propagation** (what should I see in a wavefield?)
- What is in the **engine** of simulation algorithms?
- What are the major parameters, where are **dangers** and **traps**?
- What is a good strategy **to know my simulation is right**?
- What is the difference between the various approaches, **pros and cons**?
- What are the **simplest programs** to see how things work?

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What you do not get ...

- How specific **community solvers** work (e.g., specfem, seissol, salvus, etc.)
- **Classic approaches** like ray tracing or reflectivity methods
- **Codes** for 3D solutions
- Beamforming

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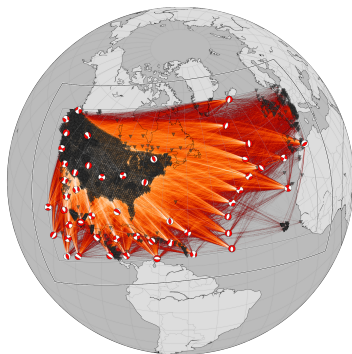
What is Computational Seismology?

We define **computational seismology** such that it **involves the complete solution of the seismic wave propagation (and rupture) problem for arbitrary 3-D models by numerical means.**

What is not covered ... but you can do tomography with ...

- **Ray-theoretical** methods
- **Quasi-analytical** methods (e.g., normal modes, reflectivity method)
- **Frequency-domain** solutions
- **Boundary integral** equation methods
- **Discrete particle** methods

These methods are important for **benchmarking** numerical solutions!



Who needs Computational Seismology

Many problems rely on the analysis of **elastic wavefields**

- **Global seismology** and tomography of the Earth's interior
- The quantification of **strong ground motion - seismic hazard**
- The understanding of the **earthquake source process**
- The monitoring of **volcanic processes** and the forecasting of eruptions
- **Earthquake early warning** systems
- **Tsunami early warning** systems
- Local, regional, and global **earthquake services**
- Global monitoring of **nuclear tests**
- **Laboratory scale analysis** of seismic events

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- **Environmental** seismology
- **Planetary seismology** - Apollo, INSIGHT
- **Exploration geophysics**, reservoir scale seismics
- **Geotechnical engineering** (non-destructive testing, small scale tomography)
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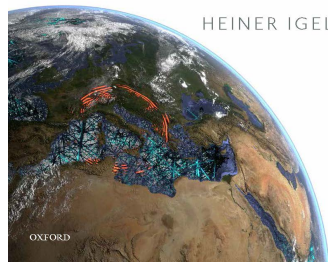
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COMPUTATIONAL SEISMOLOGY

A Practical Introduction

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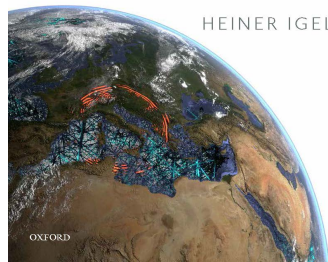


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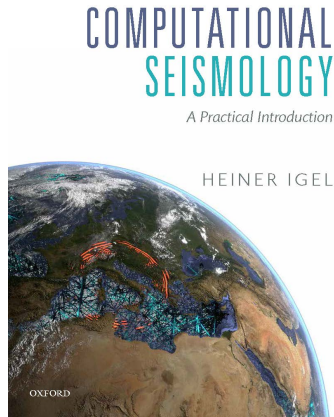
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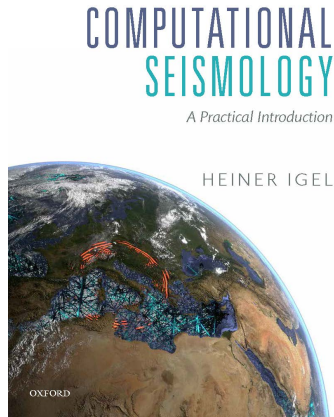
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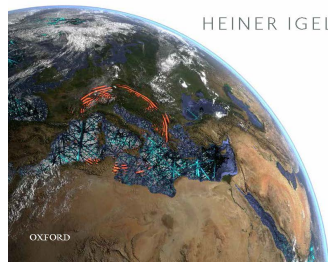


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A Practical Introduction

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9-week course in Computational wave propagation on COURSERA (free!)

Blättern > Physik und Ingenieurwesen > Forschungsmethoden

von  

Computers, Waves, Simulations: A Practical Introduction to Numerical Methods using Python

★★★★★ 4.7 289 Bewertungen

 Heiner Igel

Kostenlos anmelden
Beginnt am 26. März

Finanzielle Unterstützung verfügbar

16.370 bereits angemeldet

Covers the
finite-difference,
pseudospectral,
finite-element, and
spectral-element
method.

Why numerical methods?

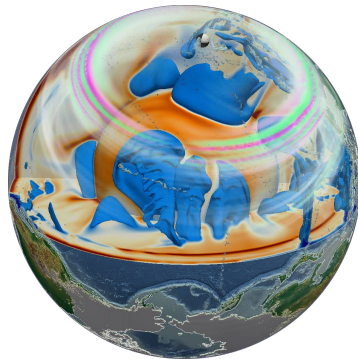


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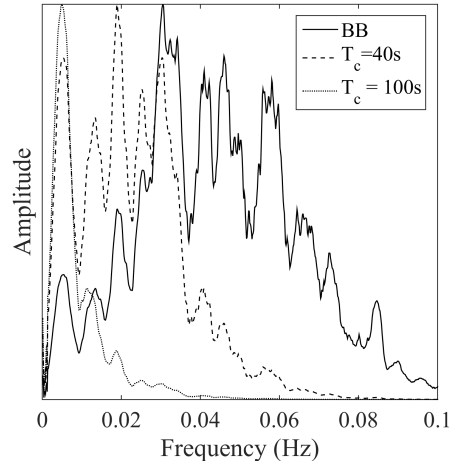
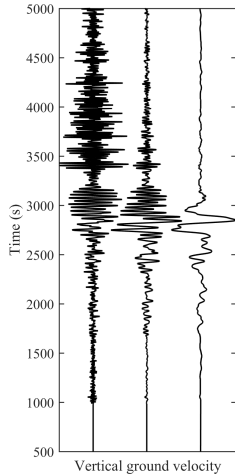
Computational Seismology, Memory, and Compute Power

Numerical solutions necessitate the **discretization** of Earth models. Estimate how much **memory** is required to store the Earth model and the required displacement fields.

Are we talking laptop or supercomputer?

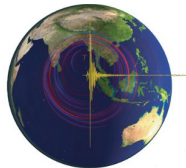


Seismic Wavefield Observations



Exercise: Sampling a global seismic wavefield

- The highest frequencies that we observe for global wave fields is 1Hz.
- We assume a homogeneous Earth (radius 6371km).
- P velocity $v_p = 10\text{km/s}$ and the v_p/v_s ratio is $\sqrt{3}$
- We want to use 20 **grid points (cells) per wavelength**
- How many grid cells would you need (assume cubic cells).
- What would be their size?
- How much memory would you need to store one such field (e.g., density in single precision).



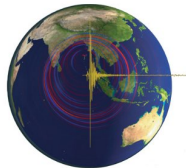
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$$c = \frac{\lambda}{T} = \lambda f = \frac{\omega}{k}$$



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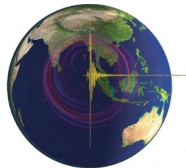
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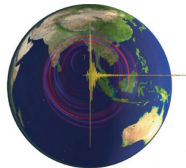
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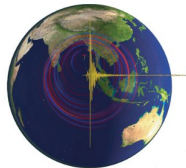
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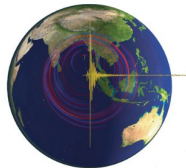
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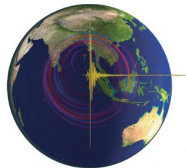
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Exercise:

```
# Physics

# Earth's Velocity (S waves have smaller wavelength)
c = 10./np.sqrt(3) # km/s

# Target Frequency
T = 1. # Hz (1 Hz is usually the highets frequency of global wave fields)

# Wavelength
lam = c * T # in km

print(' Wavelength: ', lam, ' km ')
```

Wavelength: 5.773502691896258 km

```
# Computational method
npts = 20 # number of grid points per wavelength

# Required spatial discretization
dx = lam/npts # in km

# Size of a volume cell
dv = dx**3 # in km**3

print(' Size of Volume Cell',dv, 'km**3 ')
```

Size of Volume Cell 0.024056261216234418 km**3

```
# Memory requirement (Volume Earth / dv * 8 bytes)
mem = 4./3.*np.pi*6371**3/dv*8

print(' Memory requirement : ', mem/1000/1000/1000/1000, 'TBytes')
```

Memory requirement : 360.22452769668104 TBytes

Take a few minutes **with your neighbor** to think of your current or future simulation problem (whatever physical process). What is the **physical scale** of the problem? **2D or 3D**? What are the **wavelengths** involved? How much **memory** per field (elastic parameters, displacement)? **Laptop or supercomputer**?

Exercise: Solution

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Results (@ $T = 1s$) : 360
TBytes

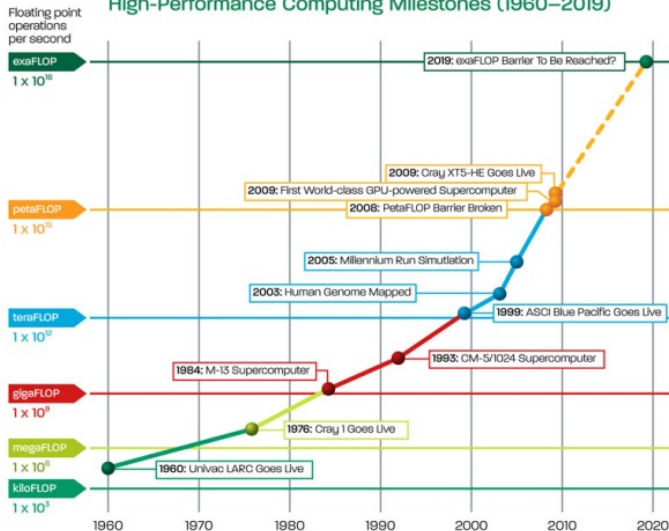
Results (@ $T = 10s$) : 360
GBytes

Results (@ $T = 100s$) : 360
MBytes

Computational Seismology, Memory, and Compute Power

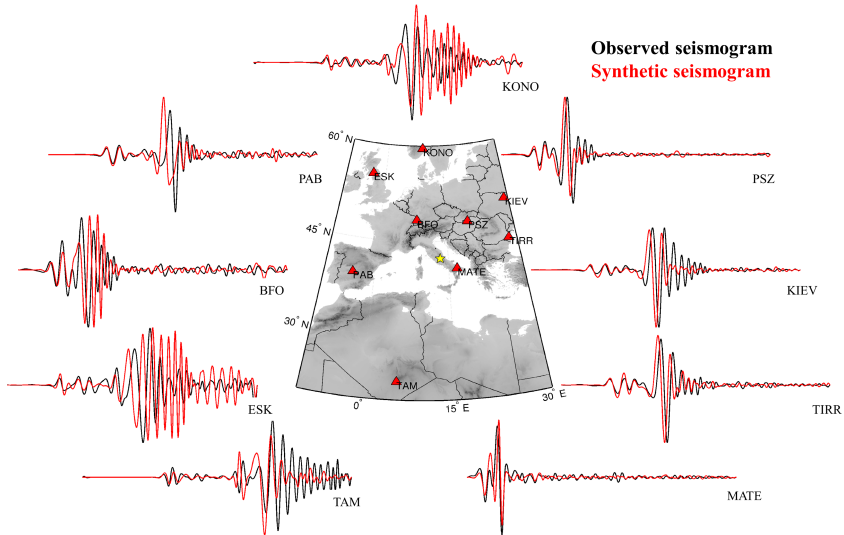


High-Performance Computing Milestones (1960–2019)



1960: 1 MFlops
1970: 10MFlops
1980: 100MFlops
1990: 1 GFlops
1998: 1 TFlops
2008: 1 Pflops
2021: 1 EFlops

The Ultimate Goal: Matching Wavefield Observations



A Bit of Wave Physics

Acoustic wave equation: no source

Acoustic wave equation

$$\partial_t^2 p = c^2 \Delta p + s$$

$p \rightarrow p(\mathbf{x}, t)$, pressure

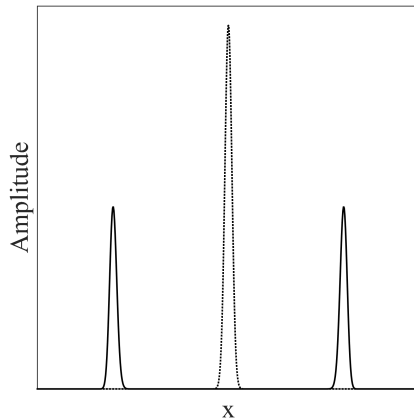
$c \rightarrow c(\mathbf{x})$, velocity

$s \rightarrow s(\mathbf{x}, t)$, source term

Initial conditions

$$p(\mathbf{x}, t = 0) = p_0(\mathbf{x}, t)$$

$$\partial_t p(\mathbf{x}, t = 0) = 0$$



Snapshot of $p(\mathbf{x}, t)$ (solid line) after some time for initial condition $p_0(\mathbf{x}, t)$ (Gaussian, dashed line), 1D case.

Acoustic wave equation: external source

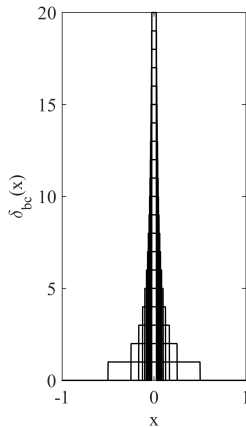
Green's Function G

$$\partial_t^2 G(\mathbf{x}, t; \mathbf{x}_0, t_0) - c^2 \Delta G(\mathbf{x}, t; \mathbf{x}_0, t_0) = \delta(\mathbf{x} - \mathbf{x}_0) \delta(t - t_0)$$

Delta function δ

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(x) dx = 1, \quad \int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0)$$



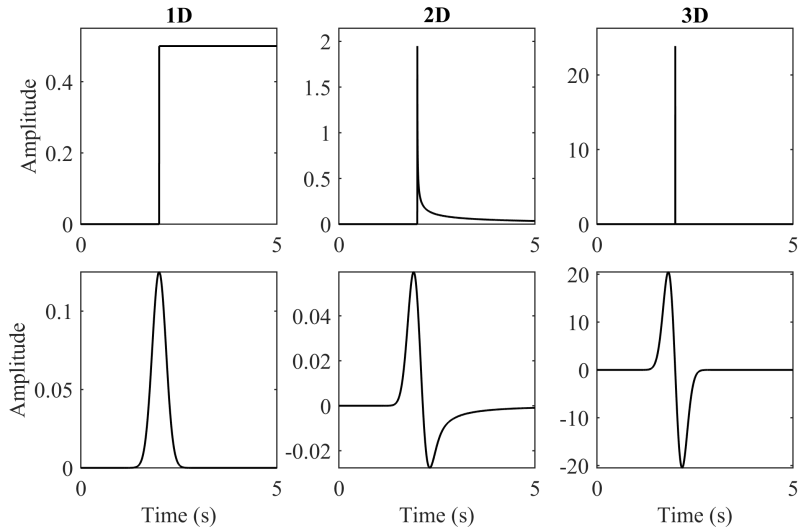
δ -generating function using boxcars.

Acoustic wave equation: analytical solutions

Green's functions for the inhomogeneous acoustic wave equation for all dimensions. $H(t)$ is the Heaviside function.

1D	2D	3D
$\frac{1}{2c} H\left(t - \frac{ r }{c}\right)$	$\frac{1}{2\pi c^2} \frac{H\left(t - \frac{ r }{c}\right)}{\sqrt{t^2 - \frac{r^2}{c^2}}}$	$\frac{1}{4\pi c^2 r} \delta(t - r/c)$
$r = x$	$r = \sqrt{x^2 + y^2}$	$r = \sqrt{x^2 + y^2 + z^2}$

Acoustic wave equation: analytical solutions



The Elastic Wave Equation

Displacement-stress formulation

$$\rho \partial_t^2 u_i = \partial_j (\sigma_{ij} + M_{ij}) + f_i$$

$$\sigma_{ij} = \lambda \epsilon_{kk} \delta_{ij} + 2 \mu \epsilon_{ij}$$

$$\epsilon_{kl} = \frac{1}{2} (\partial_k u_l + \partial_l u_k),$$

Dependencies

$$u_i \rightarrow u_i(\mathbf{x}, t) \quad i = 1, 2, 3$$

$$v_i \rightarrow v_i(\mathbf{x}, t) \quad i = 1, 2, 3$$

$$\sigma_{ij} \rightarrow \sigma_{ij}(\mathbf{x}, t) \quad i, j = 1, 2, 3$$

$$\epsilon_{ij} \rightarrow \epsilon_{ij}(\mathbf{x}, t) \quad i, j = 1, 2, 3$$

$$\rho \rightarrow \rho(\mathbf{x})$$

$$c_{ijkl} \rightarrow c_{ijkl}(\mathbf{x}) \quad i, j, k, l = 1, 2, 3$$

$$f_i \rightarrow f_i(\mathbf{x}, t) \quad i = 1, 2, 3$$

$$M_{ij} \rightarrow M_{ij}(\mathbf{x}, t) \quad i, j = 1, 2, 3,$$

1-D elastic wave equation

Shear Motion

$$\rho(x)\partial_t^2 u(x, t) = \partial_x [\mu(x)\partial_x u(x, t)] + f(x, t)$$

u displacement

f external force

ρ mass density

μ shear modulus

Velocity - Stress Formulation

Defining velocity v and stress component σ as

$$\partial_t u = v$$

$$\sigma = \mu \partial_x u$$

and assuming space-time dependencies leads to the wave equation

$$\rho \partial_t v = \partial_x \sigma + f$$

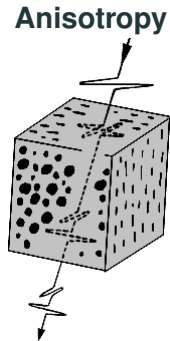
$$\dot{\sigma} = \mu \partial_x v$$

Our unknown solution vector is

$$\mathbf{q}(x, t) = (v, \sigma)$$

In order of relevance

- Viscoelasticity
- Anisotropy
- Poroelasticity
- Plasticity

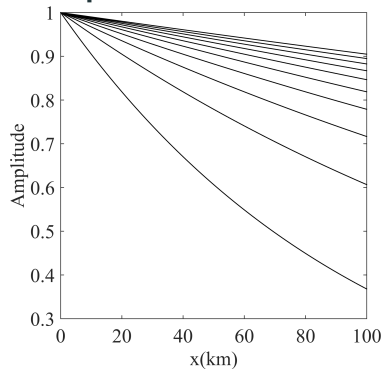


Amplitude decay

$$\frac{1}{Q(\omega)} = -\frac{\Delta E}{2\pi E}$$

$$A(x) = A_0 e^{-\frac{\omega x}{2cQ}}$$

Examples



Generalized Hooke's Law

$$\sigma_{ij} = c_{ijkl} \epsilon_{kl}, \quad i, j, k, l = 1, 2, 3$$

Reduced notation (Kelvin-Voigt)

$$c_{pq} = \begin{pmatrix} c_{11} & c_{12} & c_{13} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{13} & 0 & 0 & 0 \\ c_{13} & c_{13} & c_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{c_{11} - c_{12}}{2} \end{pmatrix}$$

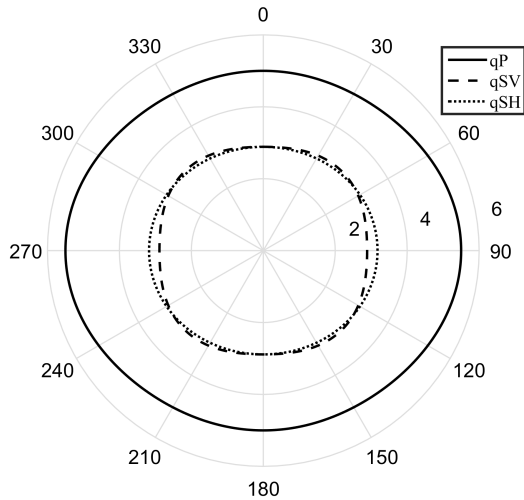
Velocity variations (Thomson parameters)

$$v_{qP}(\theta) = v_{P0} \left(1 + \delta \sin^2(\theta) \cos^2(\theta) + \epsilon \sin^4(\theta) \right)$$

$$v_{qSV}(\theta) = v_{S0} \left(1 + \frac{v_{P0}^2}{v_{S0}^2} (\epsilon - \delta) \sin^2(\theta) \cos^2(\theta) \right)_{(1)}$$

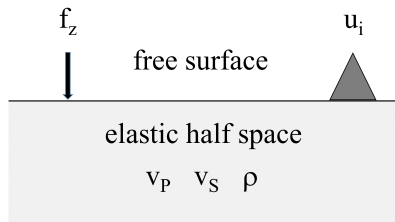
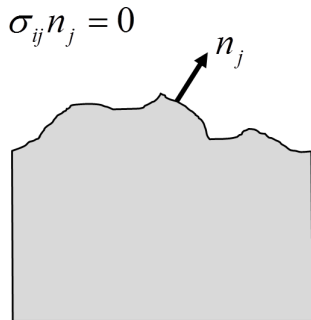
$$v_{qSH}(\theta) = v_{S0} \left(1 + \gamma \sin^2(\theta) \right)$$

Anisotropic velocities

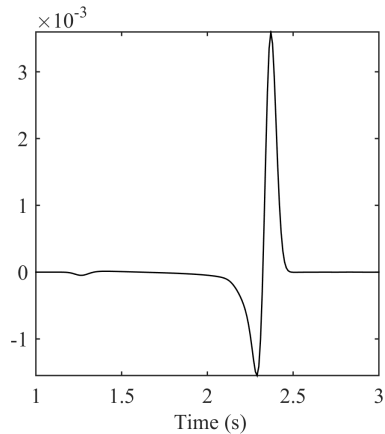
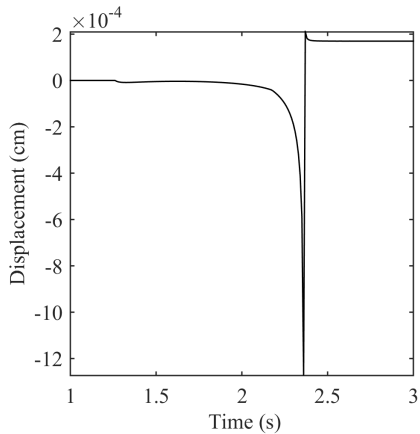


Free Surface Boundary Conditions

$$t_i = \sigma_{ij} n_j \rightarrow \sigma_{xz} = \sigma_{yz} = \sigma_{zz} = 0$$



Lamb's Problem

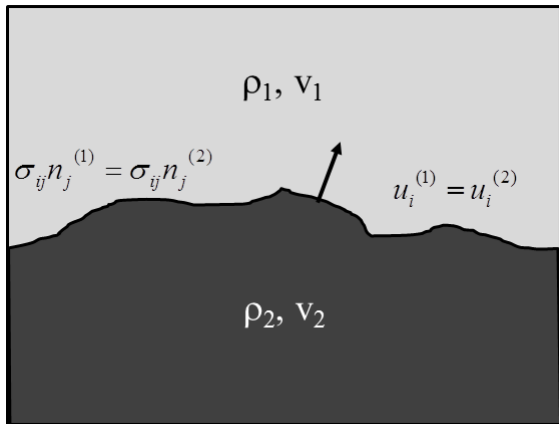


Internal Boundary Conditions

$$\sigma_{ij} n_j^{(1)} = \sigma_{ij} n_j^{(2)}$$

$$u_i^{(1)} = u_i^{(2)}$$

Internal boundary conditions need not be modelled explicitly!



Gradient, Divergence, Curl

Gradient

$$\nabla \mathbf{u}(\mathbf{x}, t) = \partial_j u_i(\mathbf{x}, t) .$$

Deformation

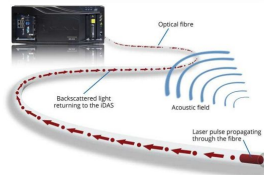
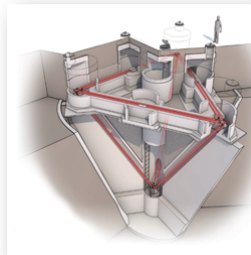
$$\epsilon_{ij}(\mathbf{x}, t) = \frac{1}{2}(\partial_i u_j(\mathbf{x}, t) + \partial_j u_i(\mathbf{x}, t))$$

Curl

$$\frac{1}{2} \nabla \times \mathbf{u} = \frac{1}{2} \begin{pmatrix} \partial_y u_z - \partial_z u_y \\ \partial_z u_x - \partial_x u_z \\ \partial_x u_y - \partial_y u_x \end{pmatrix}$$

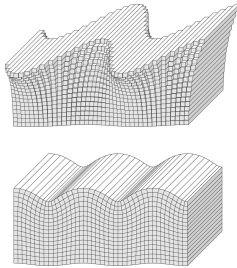
Divergence

$$\nabla \cdot \mathbf{u} = \partial_x u_x + \partial_y u_y + \partial_z u_z$$

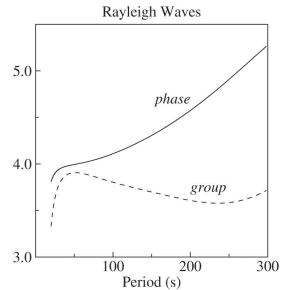
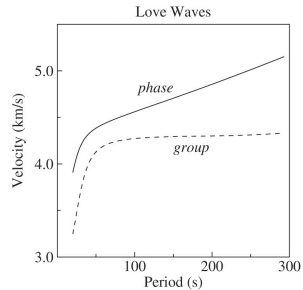


Surface Waves - Dispersion

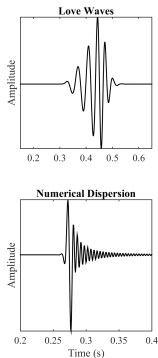
Ralyeigh and Love Waves



Dispersion Curves



Numerical Dispersion



Numerical and physical dispersion can
be confused!

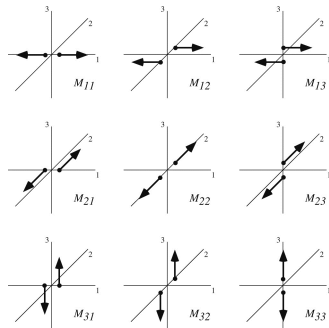
In wave simulations we have to **avoid**
numerical dispersion!

The Moment Tensor

Moment Tensor

$$\mathbf{M} = \begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix}$$

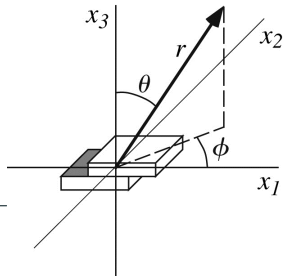
$$M_0 = \mu A d$$



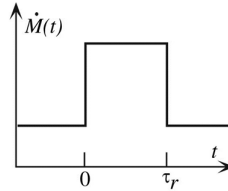
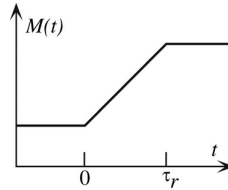
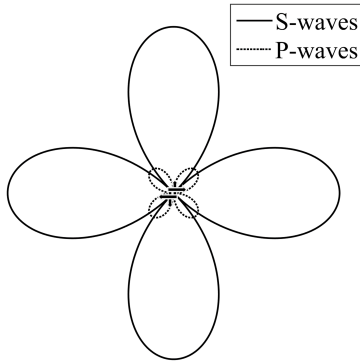
The DC analytical solution

Double Couple Green's function

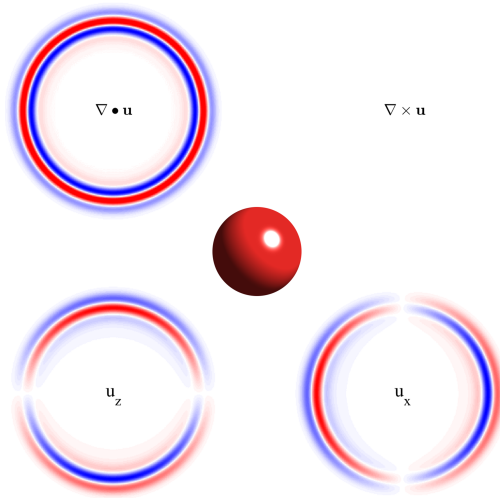
$$\begin{aligned} u(\mathbf{x}, t) = & \frac{1}{4\pi\rho} \mathbf{A}^N \frac{1}{r^4} \int_{r/\alpha}^{r/\beta} \tau M_0(t - \tau) d\tau + \\ & + \frac{1}{4\pi\rho\alpha^2} \mathbf{A}^{IP} \frac{1}{r^2} M_0(t - \frac{r}{\alpha}) + \frac{1}{4\pi\rho\beta^2} \mathbf{A}^{IS} \frac{1}{r^2} M_0(t - \frac{r}{\beta}) + \\ & + \frac{1}{4\pi\rho\alpha^3} \mathbf{A}^{FP} \frac{1}{r} \dot{M}_0(t - \frac{r}{\alpha}) + \frac{1}{4\pi\rho\beta^3} \mathbf{A}^{FS} \frac{1}{r} \dot{M}_0(t - \frac{r}{\beta}) \end{aligned}$$



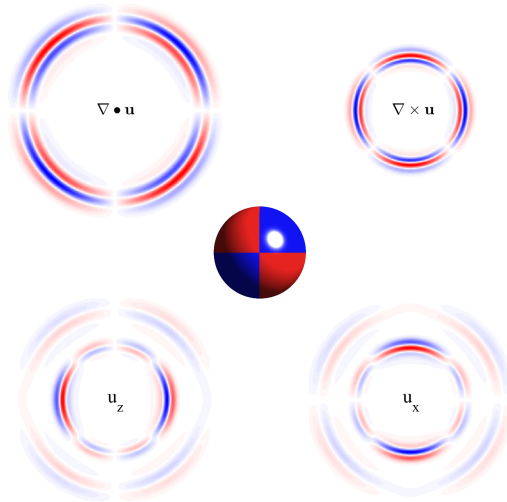
Radiation and Source Time Function



Wavefields from Moment Tensor Sources



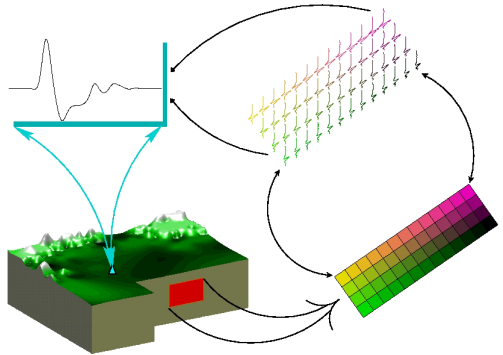
Wavefields from Moment Tensor Sources



Superposition Principle

$$v_j^r(\omega) = \sum_{k=1}^N \text{slip}_k \exp[-i\omega t_k(c^{rup})] G_{kl}^r(\omega) S(R, \omega)$$

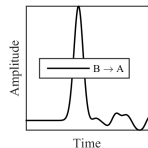
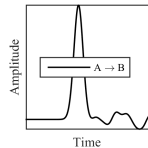
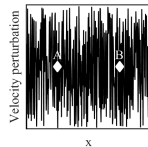
Finite sources can be simulated by summing up over point sources



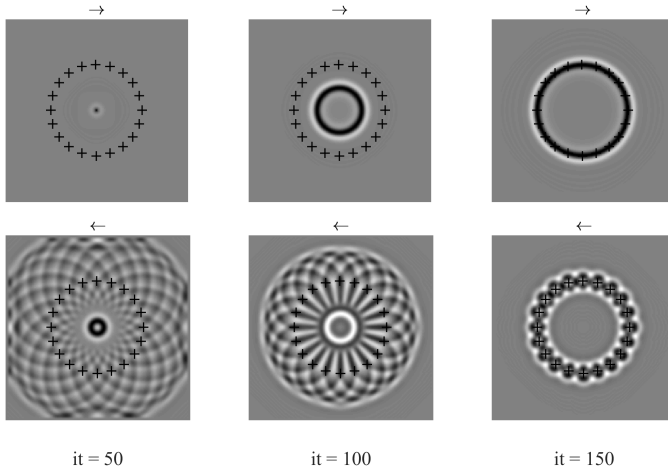
Reciprocity

$$G_{ij}(\mathbf{x}, t; \mathbf{x}_0, t_0) = G_{ji}(\mathbf{x}_0, -t_0; \mathbf{x}, -t)$$

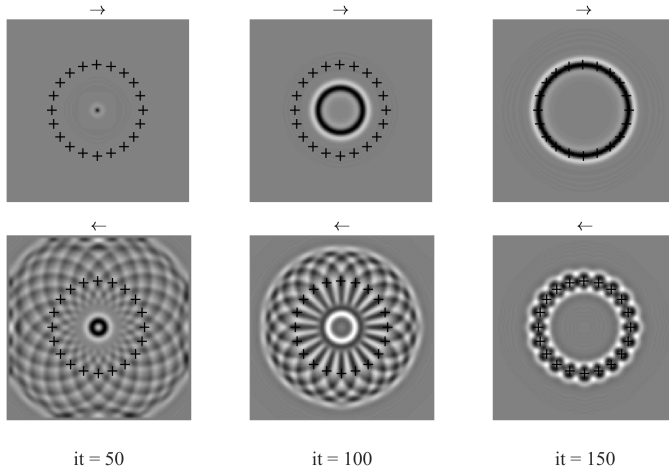
The wave equation is symmetric in time. Source and receiver locations can be interchanged. This has dramatic consequences for modeling and inversion!



Time Reversal



Time Reversal - Exercise



On github run the Jupyter Notebook 03_time_reversal_reciprocity and change the number of receivers.

Wave Equation as Linear System

Seismogram for arbitrary source $s(t)$ as convolution (exact)

$$p(\mathbf{x}, t) = G(\mathbf{x}, t, \mathbf{x}_0) \otimes s(t)$$

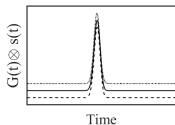
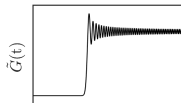
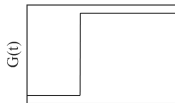
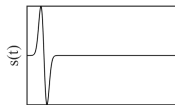
Seismogram for arbitrary source $s(t)$ as convolution (numerical)

$$\tilde{p}(\mathbf{x}, t) = \tilde{G}(\mathbf{x}, t, \mathbf{x}_0) \otimes s(t)$$

Important consequence:

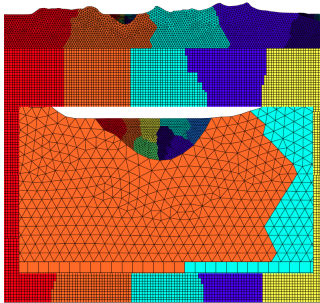
Even if your numerical Green's function $\tilde{G}(\mathbf{x}, t, \mathbf{x}_0)$ is inaccurate, the numerical solution $\tilde{p}(\mathbf{x}, t)$ might be very accurate provided the $s(t)$ is defined in the right frequency band!

Wave Equation as Linear System



- Accurate Green's functions cannot be calculated numerically
- A numerical solver is a **linear system**
- The convolution theorem applies
- Inaccurate simulations can be filtered afterwards
- Source time functions can be altered afterwards
- ... provided the sampling is good enough ...

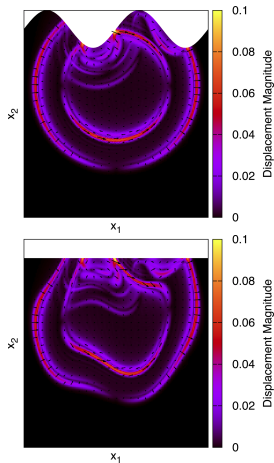
Challenges - Meshing



Human time	Simulation workflow	cpu time
15%	Design	0%
80% (weeks)	Geometry creation, meshing	10%
5%	Solver	90%

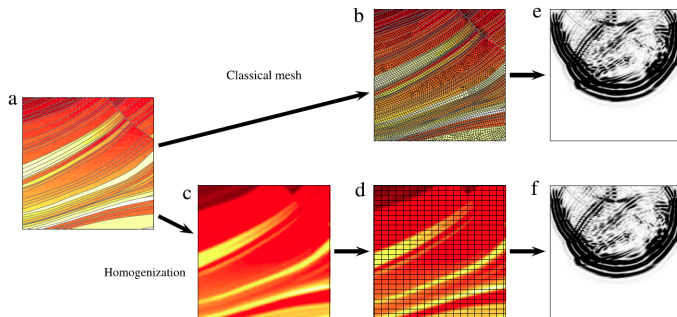
- Meshing work flow not well defined
- Still major bottleneck for simulation tasks with complex geometries
- Tetrahedral meshes easier, but ...
- Salvus?

Future Strategies - Alternative Formulations



- Particle relabelling, grid stretching
- Mapping geometrical complexity onto regular grids
- Smart pre-processing rather than meshing?
- Similar concept used in summation-by-parts (SBP) algorithms (SW4)

Future Strategies - Homogenization



- We only see low-pass filtered Earth
- So why simulate models with infinite frequencies?
- Homogenisation of discontinuous model
- Renaissance of regular grid methods?

Questions?